

Odraz a lom EM vln na rozhraní

- rovinné monochromatické vlny na rozhraní dvou prostředí s reálnými indexy lomu n_1 a n_2

dopadající $\tilde{E} = \tilde{E}_0 e^{i(\omega t - \vec{k} \cdot \vec{r})}$

odrazená $\tilde{E}_R = \tilde{E}_0 e^{i(\omega_R t - \vec{k}_R \cdot \vec{r})}$

lomená $\tilde{E}_T = \tilde{E}_0 e^{i(\omega_T t - \vec{k}_T \cdot \vec{r})}$

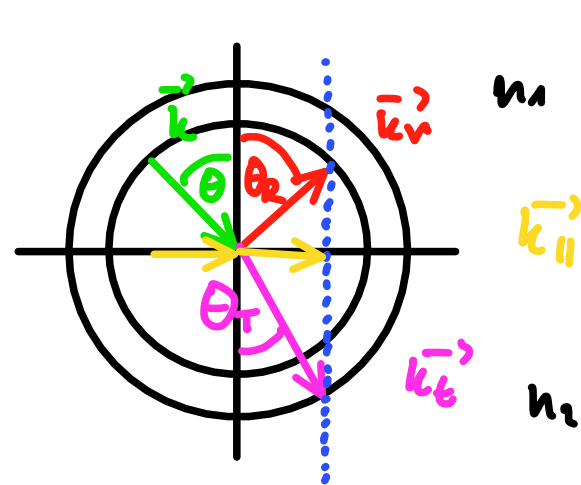
→ v rovině xy existují vlny $\Rightarrow$ rovnají se

$$\vec{r} = 0 \Rightarrow \omega_T = \omega_R = \omega$$

$$t = 0 \Rightarrow \vec{r} \cdot \vec{k}_R = \vec{r} \cdot \vec{k}_T = \vec{r} \cdot \vec{k}$$

$\vec{r}$ je v rovině $xy \Rightarrow$ zachovávají se tečné složky $\vec{k}$

$$+ \quad k = \frac{\omega}{c} = \frac{\omega}{c_0} n \quad \text{dostaneme}$$



Zákon odrazu $\theta = \theta_R$

Zákon lomu $n_1 \sin \theta = n_2 \sin \theta_T$

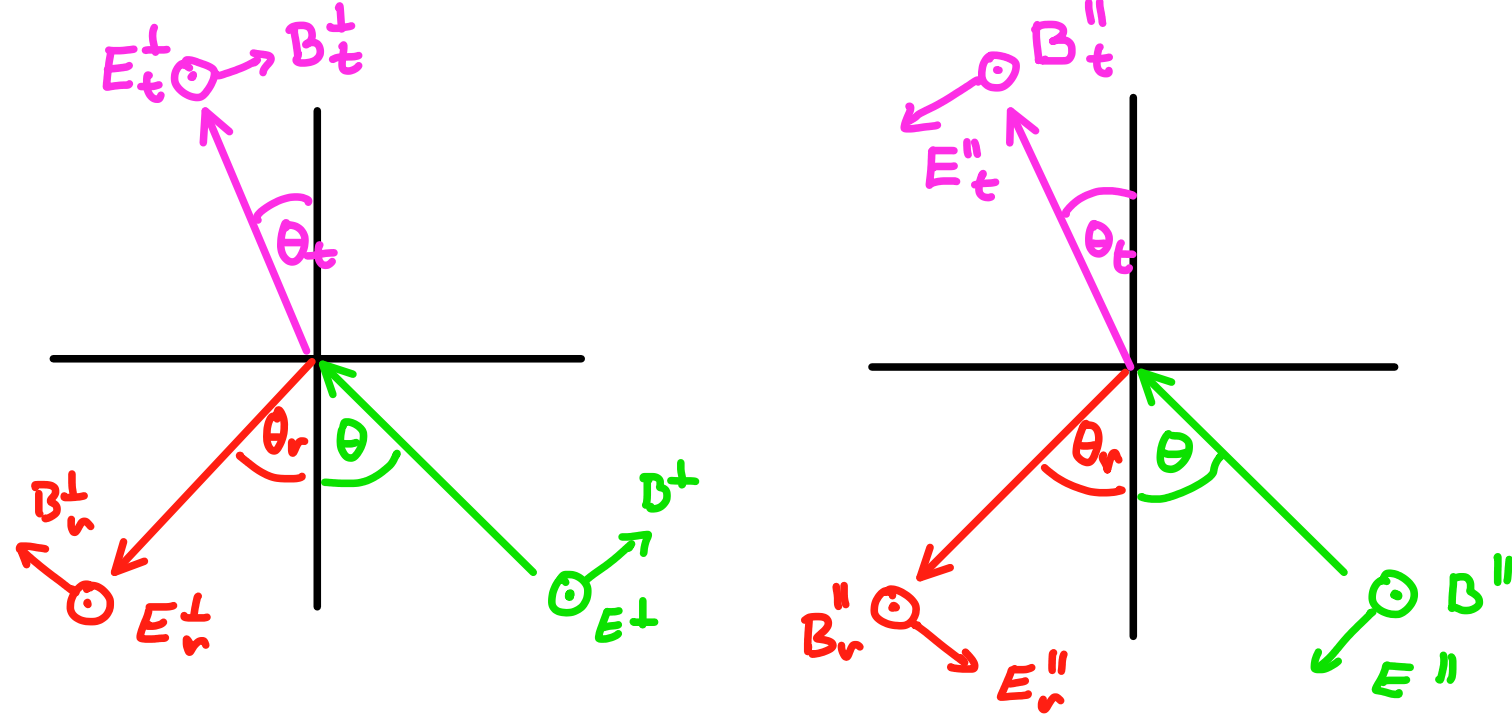
Fresnelovy vzorce

→ chceme vědět situaci z hlediska amplitud

- rozlišujeme dva významné případy

$\vec{E} \perp$ rovina dopadu

$\vec{E} \parallel$ rovina dopadu



→ z Maxwellových rovnic máme podstatně pro spojitost tečných složek k rozhraní, protože uvažujeme vakuum nikoli a proudy

$$E^\perp + E_r^\perp = E_t^\perp$$

$$B^\perp \cos \theta - B_r^\perp \cos \theta_r = B_t^\perp \cos \theta_t$$

$$B^\parallel + B_r^\parallel = B_t^\parallel$$

$$E^\parallel \cos \theta - E_r^\parallel \cos \theta_r = E_t^\parallel \cos \theta_t$$

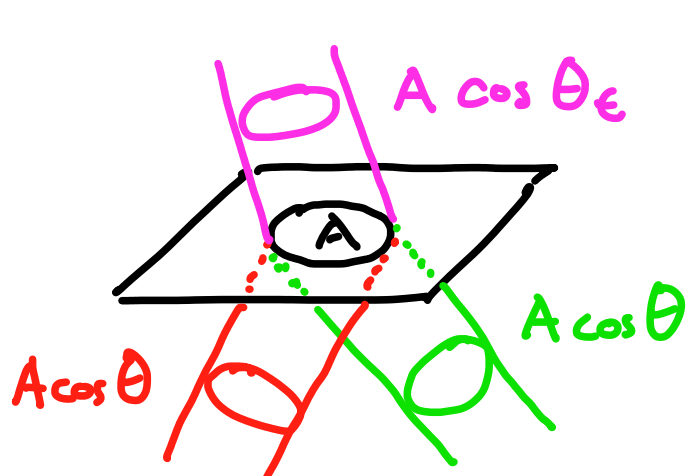
→ použijeme $E = cB = \frac{c_0}{n} B$ + $\theta = \theta_t$

$$r^\perp := \frac{E_r^\perp}{E^\perp} = \frac{n_1 \cos \theta - n_2 \cos \theta_t}{n_1 \cos \theta + n_2 \cos \theta_t}$$

$$r^\parallel := \frac{E_r^\parallel}{E^\parallel} = \frac{n_2 \cos \theta - n_1 \cos \theta_t}{n_2 \cos \theta + n_1 \cos \theta_t}$$

$$t^\perp := \frac{E_t^\perp}{E^\perp} = \frac{2n_1 \cos \theta}{n_1 \cos \theta + n_2 \cos \theta_t}$$

$$t^\parallel := \frac{E_t^\parallel}{E^\parallel} = \frac{2n_1 \cos \theta}{n_2 \cos \theta + n_1 \cos \theta_t}$$



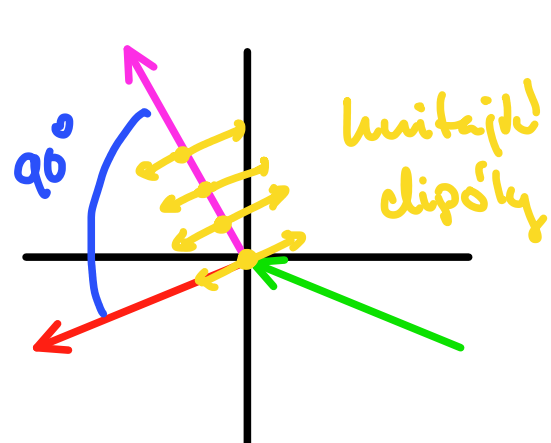
$$J = \vec{S} \cdot \vec{A} = n \frac{1}{2} c_0 \epsilon_0 |E|^2$$

$$R^{\perp/\parallel} = \frac{J_r^{\perp/\parallel}}{J^{\perp/\parallel}} = (r^{\perp/\parallel})^2$$

$$T^{\perp/\parallel} = \frac{J_t^{\perp/\parallel}}{J^{\perp/\parallel}} = (t^{\perp/\parallel})^2 \frac{n_2 \cos \theta_t}{n_1 \cos \theta}$$

Brewsterův úhel

$$\rightarrow r^\parallel = 0 \Leftrightarrow \tan \theta_B = \frac{n_2}{n_1}$$



$\Rightarrow$ polarizace odrazem

Kritický úhel

$$\rightarrow \theta_c \leftrightarrow \theta_t = \frac{\pi}{2} \xrightarrow{\text{Snell}} \sin \theta_c = \frac{n_2}{n_1}$$